

Section 11: Conflicting & Noisy Data — Hallucination Detection

■ **LangGraph Node: ValidationNode** — Detects when two nodes return contradicting facts. The graph must route through this node before synthesis when multiple sources are retrieved.

The following records from a SECONDARY SOURCE (unofficial cricket stats portal) conflict with the primary data in Sections 2–3. Students must implement a ValidationNode that detects and flags these conflicts rather than silently returning one value.

Player / Fact	Primary Source Value	Secondary Source Value	Conflict Type	Expected RAG Behaviour
Virat Kohli — Career Runs	7263	7084	Value mismatch	Flag conflict, cite both, suggest verification
Yuzvendra Chahal — Career Wickets	205	187	Value mismatch (retired data?)	Flag — secondary may be older dataset
MI vs CSK — Total Matches	35	33	Count mismatch	Flag conflict, both sources cited
Highest Team Score	287/3 (SRH, 2024)	263/5 (RCB, 2013)	Different records by year	Clarify: 287/3 is current record; 263/5 was previous
MS Dhoni — Matches Played	250	240	Count mismatch	Flag — 250 is from official IPL; secondary is incomplete
Best Bowling Figures	6/12 (Alzarri Joseph)	6/14 (Sohail Tanvir)	Different records	6/12 is correct; 6/14 was old record. Explain superseded.

ValidationNode Implementation Hint: After retrieval, compare numeric values for the same entity across chunks. If the same field (e.g., 'career runs') has two different values from two different sources, set a flag in state: state['conflict_detected'] = True. Route to ValidationNode which builds a conflict-aware prompt: 'Two sources disagree on X. Source A says Y. Source B says Z. Please flag this to the user and recommend verification.'